

Computational Social Science with Images and Audio

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06 October 2023

CNN are the current computer vision workhorse.

But not the only horse in the stable!

- ▶ Convolutional neural nets (abbreviated as CNNs or CovNets) are specialized neural networks for CV tasks
 - ▶ For various tasks, such as object detection, facial recognition, image segmentation, sentiment analysis
- ▶ Much of the recent image-based social science research uses CNN
- ▶ Other approaches are used, too:
 - ▶ Classic techniques: in the pre-deep learning and pre-CNN era, CV relied on hand-crafted features and algorithms
 - ▶ Edge detection, color histograms, template matching, ...
 - ▶ Other neural nets: recurrent neural networks (RNN), generative adversarial networks (GAN), transformers, stable diffusion models
 - ▶ RNN are often used for video analysis, GAN for image generation
 - ▶ Hybrid approaches: traditional approaches and deep learning
 - ▶ Unsupervised and semi-supervised learning: e.g., clustering

One promise of neural nets is a more end-to-end approach.

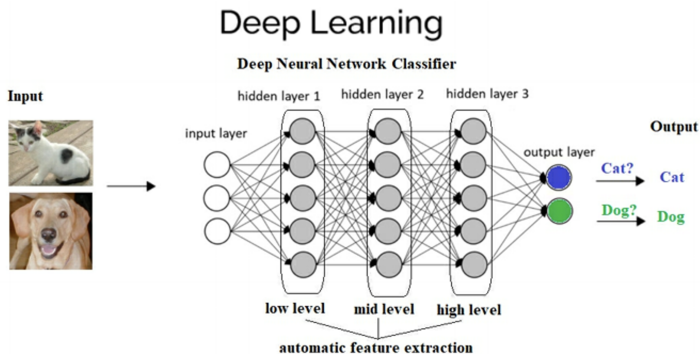


Figure: Dey (2018)¹

¹Dey, S. (2018). Hands-On Image Processing with Python: Expert Techniques for Advanced Image analysis and Effective Interpretation of Image Data. Packt Publishing Ltd.

CNN typically contain an input layer, hidden layers, and an output layer.

- ▶ Typically: input layer → hidden layers → output layer
- ▶ What do different convolutional layers do?
 - ▶ Layers close to the input layer learn low-level features (e.g., lines)
 - ▶ Middle layers learn complex abstract features (combining lower-level features)
 - ▶ Layers closer to the output interpret the extracted features in the light of the classification task

A typical application for a CNN in social science could be a visual sentiment analyzer.

- ▶ Two ways to think about emotions and images: expressed and evoked
 - ▶ What emotions are expressed? → Typically facial analysis/emotion annotation, objective-ish
 - ▶ What emotions are evoked? → For example, image sentiment, subjective-ish
- ▶ CNNs do a moderately good job of predicting evoked emotions (60–70% accuracy),² depending on...
 - ▶ The dataset being used
 - ▶ The architecture of the CNN
 - ▶ The training process
 - ▶ Maybe most importantly: How “evoked emotions” are defined and labeled

²Peng, K.-c., et al. (2015). A Mixed Bag of Emotions: Model, Predict, and Transfer Emotion Distributions. In: 2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR) pp. 1–9.

What emotions do these images evoke?

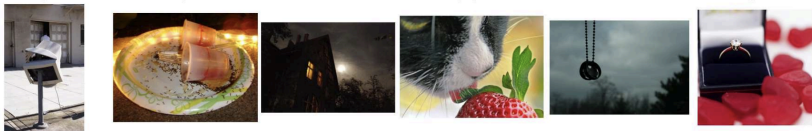


Figure: Source: Peng, K.-c., et al. (2015)

What emotion does each of these images evoke?

“Ground truth” in the Emotion6 Database (focusing on Ekman's 6 basic emotions plus neutral)

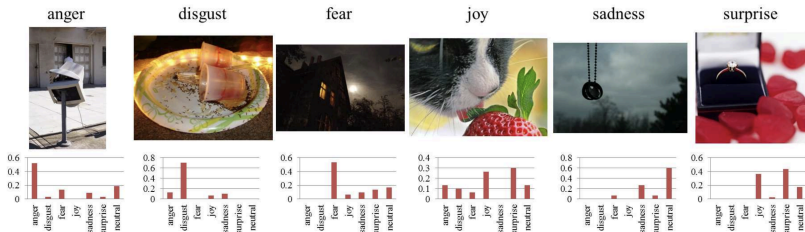


Figure: Source: Peng, K.-c., et al. (2015)

CNN are traditionally employed in supervised settings.

- ▶ CNNs are a specialized type of deep learning algorithm working with raw pixel values
- ▶ CNNs use many prelabeled training images to “learn” which pixel combinations are associated with the desired labels
- ▶ Once the performance is satisfactory, the algorithm can then be used to label large numbers of additional images quickly and at low cost
- ▶ Existing trained CNNs can be borrowed and fine-tuned (“transfer learning”) for (rather) accurate results using a smaller training set of images
 - ▶ When is this a good/bad idea?
- ▶ Can you think of social science-relevant examples other than sentiment?

Training CNN requires substantial computing power and a lot of data.

- ▶ Access to graphics processing unit (GPU) *greatly* increases efficiency
- ▶ Supervised learning algorithms like CNNs require labeled images:
 - ▶ Options for generating original labels include hiring coders, using commercial services like LabelBox, Mechanical Turk
 - ▶ Researchers can also use benchmark image datasets with well-validated labels
 - ▶ Or rely on “creative” ways of labeling
 - ▶ For instance, annotate partisanship based on the data point’s source
 - ▶ Intuition: a desired label might strongly correlate with an observable characteristic of the data (“proxy variable”)
 - ▶ What are some ethical challenges of image annotation?
- ▶ A lot of data requires a lot of (robust) storage

Data labeling still represents *a lot* of human work.

'It's destroyed me completely': Kenyan moderators decry toll of training of AI models

Employees describe the psychological trauma of reading and viewing graphic content, low pay and abrupt dismissals



📍 Office premises of Sama in Nairobi, Kenya. Photograph: Tony Karumba/AFP/Getty Images

Figure: Source: The Guardian (2023)³

³ "It's destroyed me completely": Kenyan moderators decry toll of training of AI models. The Guardian. (2023). <https://www.theguardian.com/technology/2023/aug/02/ai-chatbot-training-human-toll-content-moderator-meta-openai>

Von Kempelen's "automaton": a technology masterpiece?

- ▶ In 1769, inventor Wolfgang von Kempelen introduced a chess-playing automaton⁴
- ▶ It was a robot/“dummy” in Turkish attire, seated next to a large cabinet with machinery and a chessboard on top
- ▶ Visitors could inspect the cabinet and its complex machinery
- ▶ The Turk defeated notable figures such as Napoleon Bonaparte and Benjamin Franklin
- ▶ The term “automaton” gained popularity from this narrative
- ▶ Despite von Kempelen's renowned inventor status, skepticism persisted due to the machine's unbelievable capability
- ▶ Have you heard the story? What do you think?

⁴Story as recounted in: Shiller, R. J. (2019). Narratives about technology-induced job degradations then and now (No. w25536). National Bureau of Economic Research.

Von Kempelen's "automaton": an illusion masterpiece!

- ▶ There was a complex mechanism that obscured the visitors':
- ▶ They did not see that there was a man, a chess master, hiding inside
- ▶ A magnet system allowed the master to see the positions of the pieces on the board
- ▶ A system of controls enabled him to move the arms of the automaton to move the pieces

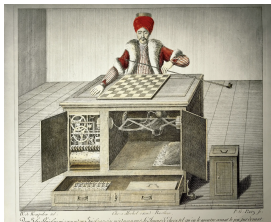


Figure: Source: Encyclopaedia Britannica, Photo credits: INTERFOTO/Alamy

Getting back to the technicalities: CNN are ANN.

- ▶ Artificial Neural Networks (ANN) are a broad category of statistical models
- ▶ They use **intermediate abstract representations** (hidden layers) of the input data to learn and predict outcomes
- ▶ CNN are an ANN subtype for processing grid-like data structures such as images, where spatial hierarchies and local patterns (edges, textures, etc.) are important
- ▶ These intermediate representations result from complex interactions and nonlinear transformations of the input matrix
- ▶ Coefficients are estimated for these intermediate representations
- ▶ *Deep* learning refers to the presence of *multiple* hidden layers
- ▶ Less standardized notions of “breadth” (e.g., # of neurons in a layer)

Convolution is fundamental for the abstract representations in CNN.

- ▶ The primary operation in CNNs is convolution, not just simple matrix multiplication
- ▶ Convolution involves applying a filter (or kernel) over the input data to produce a feature map
- ▶ This helps detect local patterns, such as edges or textures, in different regions of the input data
- ▶ Along with convolution, non-linear transformations are applied using an activation function (e.g., RELU) to introduce non-linearity into the model
 - ▶ This helps the network capture complex relationships in the data
- ▶ These feature maps (or convolved outputs) form new abstract representations that can be fed into subsequent layers

Nonlinear transformations are commonly applied to the hidden layers in deep learning models.

- ▶ Nonlinear transformations are also known as activation functions
- ▶ Common activation functions include RELU, Sigmoid, and Tanh

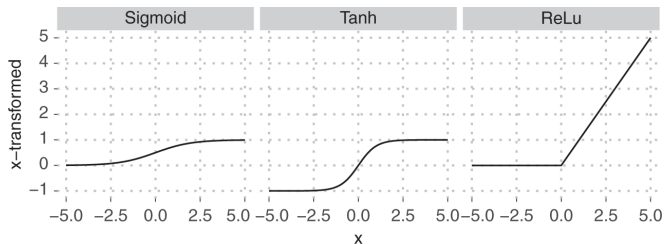


Figure: Three common nonlinear transformations (activation functions) in deep learning. Source: Webb Williams et al. (2020), cf. first slide set.